

TEST REPORT UL 2271 Standard for Safety for Batteries for Use In Light Electric Vehicle (LEV) Applications	
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Manufacturer's name.....:	Zhuhai Pengsen Electronic Technology Co., Ltd
Address.....:	2nd Floor, Building A3, No.6 Sixth Road, Xinqing Science and Technology Industrial Park, Jing'an Town, Doumen District, Zhuhai City
Test specification:	
Standard.....:	UL 2271:2018
Test procedure.....:	Type test
Non-standard test method.....:	N/A
Test Report Form No.....:	UL 2271
Test Report Form(s) Originator.....:	UL(US)
Master TRF.....:	2013-11
Test item	
Description	Lithium Ion Battery
Trade Mark	N/A
Model/Type reference	YG7455-RJ2302
Model differences.....:	--
Ratings	74.4V, 55Ah, 4092Wh



Copy of marking plate:

Lithium Ion Battery
Model: YG7455-RJ2302
Rated capacity: 74.4V, 55Ah, 4092Wh
DD/MM/YYYY
Zuhai Pengsen Electronic Technology Co., Ltd
Made in China



Test item particulars:	
Information about the product needed to establish a correct test program, such as product mobility, type of power connections and similar.	(Test item particulars are selected by the TRF Originator base on the requirements in the standard)
Designation.....:	Lithium Ion Battery
Trademark.....:	N/A
Nominal voltage..... :	74.4 V
Rated capacity..... :	55Ah
Maximum charge voltage..... :	85V
Maximum charge current.....:	27.5A
Final voltage..... :	52V
Max Ambient Temperature.....:	55°C max (charge), 60°C max (discharge)
Manufacturer's charge method.....:	Charging the battery with 11A constant current and 85V constant voltage until the current less than 1.1A at ambient 20°C±5°C
Sample No..... :	001#~017#
Possible test case verdicts:	
Test case does not apply to the test object.....: N(/A)	
Test object does meet the requirement.....: P(ass)	
Test object does not meet the requirement.....: F(ail)	
Testing:	
Date of receipt of test item.....:2025-06-30	
Date(s) of performance of tests.....: 2025-06-30 to 2025-07-02	
General remarks:	
This report shall not be reproduced, except in full, without the written approval of the testing laboratory. The test results presented in this report relate only to the object tested.	
"(see remark #)" refers to a remark appended to the report. "(see appended table)" refers to a table appended to the report.	
Throughout this report a point is used as the decimal separator.	

Summary of testing:

The component cell is approved by UL according to UL 1642.

Charging method declared by the manufacturer in specification: Charging the battery with 11A constant current, then 85V constant voltage until current reduces to 1.1A at ambient 20°C±5°C.

Test items:

cl.23 Overcharge Test;
cl.24 Short Circuit Test;
cl.25 Overdischarge Test;
cl.26 Temperature Test; cl.27 Imbalanced Charging Test.; cl.30 Vibration Endurance Test;
cl.31 Shock Test; cl.32 Crush Test; cl.31 Shock Test; cl.33 Drop Test; 36 Roll Over Test;

Description of the product:

The battery is constructed with lithium-ion cell, and has overcharge, overdischarge, over current and short-circuits protection circuit.

The battery mainly composed of:

- Protective circuit module
- Battery cell
- Plastic case

- (1) The battery has not been evaluated in combination with charger(s) or host product(s). Additional evaluation to determine compliance may be required on the combination(s) in the end product evaluation.
- (2) The battery was evaluated for a maximum charger current and maximum voltage limit outlined in the table below. The end product evaluation shall ensure that current and voltage limits noted are maintained.

The main features of the battery is shown as below:

Model	Nominal capacity	Nominal voltage	Nominal Charge Current	Nominal Discharge Current	Max. Charge Current	Max. Discharge Current	Max. Charge Voltage	Cut-off Voltage
TCRFD06 1304HYJ	55Ah	74.4V	11A	27.5A	27.5A	55A	85V	52V

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict

INTRODUCTION			
1	Scope		P
1.1	These requirements cover electrical energy storage assemblies (EESAs) such as battery packs and combination battery pack-electrochemical capacitor assemblies and the subassembly/modules that make up these assemblies for use in light electric-powered vehicles (LEVs) as defined in this standard.		P
1.2	This standard does not evaluate the performance or reliability of these devices.		P
1.3	This standard does not include requirements for the evaluation of EESAs intended for use in electric vehicles, such as on-road passenger vehicles intended for use on public roadways including highways and heavy duty off road vehicles such as battery powered ride-on industrial trucks, which are covered under the Standard for Batteries for Use in Electric Vehicles, UL 2580 / ULC-S2580.		P
1.4	This standard does not include requirements for evaluation of EESAs intended for use in light electric rail (LER) applications, which are covered under the Standard for Batteries for Use in Light Electric Rail (LER) Applications and Stationary Applications, UL 1 973.		P
2	Components		P
2.1	Except as indicated in 2.2, a component of a product covered by this Standard shall comply with the requirements for that component. See Annex A for a list of Standards covering components generally used in the products covered by this Standard. A component shall comply with the Standards of CSA or UL Standards as appropriate for the country where the product is to be used.		P
2.2	A component is not required to comply with a specific requirement that: a) Involves a feature or characteristic not required in the application of the component in the product covered by this standard; or b) Is superseded by a requirement in this standard.		P
3	Units of Measurement		P
3.1	Values stated without parentheses are the requirement. Values in parentheses are explanatory or approximate information. The values given in SI (metric) units shall be normative.		P
4	Undated References		P
4.1	Any undated reference to a code or standard appearing in the requirements of this standard shall be interpreted as referring to the latest edition of that code or standard.		P
5	Reference Publications		P
5.1	Throughout this standard, the CSA and ULC standard references apply to products intended for use in Canada, while the UL standard references apply to products intended for use in the United States. Combined references are commonly separated by a slash (" / ").		P
6	Glossary		-

UL2271			
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CONSTRUCTION			
7	Non-Metallic Materials		P
7.1	The materials employed for enclosures shall comply with the applicable enclosure requirements outlined in the Evaluation of Properties of Polymeric Materials, CAN/CSA-C22.2 No. 0.1 7 / Standard for Polymeric Materials - Use in Electrical Equipment Evaluations, UL 746C, Path III of the Enclosure Requirements Table, except as modified by this standard.	Plastic case used.	P
7.2	Polymeric materials employed for enclosures shall have a minimum flame rating of V-1 in accordance with the Evaluation of Properties of Polymeric Materials, CAN/CSA-C22.2 No. 0.1 7 / Standard for Tests for Flammability of Plastic Materials for Parts in Devices and Appliances, UL 94.		P
7.3	The following factors in (a) - (e) are taken into consideration when an enclosure employing nonmetallic materials is being judged. For a nonmetallic enclosure all of these factors are to be considered with respect to thermal aging. Dimensional stability of a polymeric enclosure is addressed by compliance to the mold stress relief test. Suitability to factors (a) - (e) below may be determined by the tests of this standard. a) Resistance to impact; b) Crush resistance; c) Abnormal operations; d) Severe conditions; and e) Mold-Stress Relief Distortion.		P
7.4	The polymeric materials employed for enclosures and insulation shall be suitable for anticipated temperatures encountered in the intended application. EESA enclosures shall have a Relative Thermal Index (RTI) with impact suitable for temperatures encountered in the application but no less than 80° C (1 76° F), as determined in accordance with the Evaluation of Properties of Polymeric Materials, CAN/CSA-C22.2 No. 0.1 7 / Standard for Polymeric Materials - Long Term Property Evaluations, UL 746B.		P
7.5	The enclosure materials intended to be directly exposed to sunlight and rain in the end use application shall comply with the UV Resistance and the Water Exposure and Immersion tests in accordance with the Evaluation of Properties of Polymeric Materials, CAN/CSA-C22.2 No. 0.1 7 / Standard for Polymeric Materials - Use in Electrical Equipment Evaluations, UL 746C.		P

UL2054			
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7.6	Materials employed as electrical insulation in the assembly shall be resistant to deterioration that would result in an electrical shock, fire or other safety hazard. Compliance is determined by the tests of this standard. Materials employed for direct support of live parts at hazardous voltage, shall additionally meet the direct support insulation criteria outlined in the Material Property Considerations table in the Evaluation of Properties of Polymeric Materials, CAN/CSA-C22.2 No. 0.1 7 / Standard for Polymeric Materials - Use in Electrical Equipment Evaluations, UL 746C unless employed as part of a component that has been evaluated to a suitable component standard. Insulated wiring is subjected to the requirements outlined in Section 1 0.	No such Materials	N/A
7.7	Gaskets and Seals of the EESA relied upon for safety, shall be determined suitable for the environmental conditions and chemical substances they are anticipated to be exposed to in their end use.		N/A
8	Metallic Parts Resistance to Corrosion		N/A
8.1	Metal EESA enclosures shall be corrosion resistant. A suitable plating or coating process can achieve corrosion resistance. Additional guidance on methods to achieve corrosion protection can be found in the Standard for Enclosures for Electrical Equipment, Environmental Considerations, CAN/CSA-C22.2 No. 94.2 / UL 50E.		N/A
8.2	Metallic EESA enclosures may be provided with an insulating liner to prevent shorting of live parts to the enclosure. If using an insulating liner for this purpose, the insulating liner shall consist of non-moisture absorbent materials that have a temperature rating suitable for temperatures within the pack during operation.		N/A
8.3	Conductive parts in contact at terminals and connections shall not be subject to corrosion due to electrochemical action. Combinations above the line in Table B.1 of Annex B shall be avoided.		N/A
9	Enclosures		P
9.1	The enclosure of an EESA shall have the strength and rigidity required to resist the possible physical abuses that it will be exposed to during its intended use, in order to reduce the risk of fire or injury to persons.		P
9.2	A tool providing the mechanical advantage of a pliers, screwdriver, or similar tool, shall be the minimum mechanical capability required to open the enclosure.		N/A

UL2271			
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9.3	Openings in the enclosure of an electric energy storage assembly shall be designed to prevent inadvertent access to hazardous parts as installed in the vehicle. Compliance is determined by the " Tests for Protection Against Access to Hazardous Parts Indicated by the First Characteristic Numeral, Clause 1 2" of the Standard for Degrees of Protection Provided by Enclosures (IP Code), IEC 60529, for a minimum IP rating of IP3X with consideration of the device as it is installed in the vehicle. Evaluation per the Standard for Degrees of Protection Provided by Enclosures (IP Code), IEC 60529, Clause 1 2, consists of the use of the 2.5 mm Test Rod, shown in Figure 1 , applied with a force of 1 0 N $\pm$ 1 0%.		P
9.4	Openings in the enclosure shall be designed to prevent ingress of water as installed in the vehicle in accordance with intended use and IP rating in accordance with Standard for Degrees of Protection Provided by Enclosures (IP Code), IEC 60529 with a minimum rating of IPX3. Compliance is determined by the Water Exposure Test (IP code rating) in Section 39.		P
10	Wiring and Terminals		P
10.1	Wiring shall be insulated and acceptable for the purpose, when considered with respect to temperature, voltage, and the conditions of service to which the wiring is likely to be subjected within the equipment.		P
10.2	Internal wiring shall be routed, supported, clamped or secured in a manner that reduces the likelihood of excessive strain on wire and on terminal connections; loosening of terminal connections; and damage of conductor insulation. In safety critical circuits, for soldered terminations, the conductor shall be positioned or fixed so that reliance is not placed upon the soldering alone to maintain the conductor in position.		P
10.3	An EESA external terminal shall be designed to prevent inadvertent shorting. An external terminal shall be designed to prevent inadvertent misalignment or disconnection when installed in its end use vehicle.		N/A
10.4	For all EESAs, an external terminal for charging shall be designed to prevent an inadvertent shorting and misalignment and a reverse polarity connection. For EESAs that are intended for removal from the vehicle for external charging or replacement with a charged EESA, the external terminal for discharging shall be designed to prevent inadvertent shorting, a reverse polarity connection and a misalignment.		N/A
10.5	For EESAs that are intended for removal from the vehicle for external charging or replacement with a charged EESA by the user, the external terminal for		N/A

UL2271			
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	discharging and any other external terminals with hazardous voltage shall be designed to prevent access by the user. Compliance is determined by use of the articulate test finger shown in Figure 2. The articulate test finger requirement does not apply to the manual disconnection for servicing device as outlined in 1 5.5.		
10.6	The external terminals of an EESA intended for removal from the vehicle for charging, shall be evaluated to either the no load endurance test or endurance with load test as applicable to the end use application in accordance with the Standard for Plugs, Receptacles and Couplers for Electric Vehicles, UL 2251 without being subjected to the exposure to contaminants.		N/A
10.7	A hole by which insulated wires pass through an internal or external opening in the enclosure shall be provided with a smoothly rounded bushing or shall have smooth surfaces, free of burrs, fins, sharp edges, and the like, upon which the wires may bear, to prevent abrasion of the insulation.		N/A
10.8	Wiring for hazardous voltage shall be readily distinguishable from low voltage wiring harnesses through the use of orange colored insulation or orange wiring harness jacketing or covers. Internal wiring of hazardous voltage circuits that are not provided with orange wiring harness covers or jackets, shall also be colored primarily orange (i.e. solid orange or orange striped), or enclosed in junction boxes with hazardous voltage warning labels such as ISO No. 701 0, No. W001 (i.e. lightning bolt within triangle).		N/A
11	Fuses		N/A
11.1	Fuses used in an EESA shall be acceptable for current and voltage which are likely to be subjected.		N/A
11.2	For user replaceable fuses, a fuse replacement marking shall be located adjacent to each fuse or fuse holder, or on the fuse holder, or in another location provided that it is obvious to which fuse the marking applies, and giving the fuse ratings. Where user replaceable fuses with special fusing characteristics such as time delay or breaking capacity are necessary, the type shall also be indicated. Information on proper fuse replacement of user replaceable fuses shall also be included in the instructions.		N/A
12	Handles		N/A
12.1	An integral handle(s) or other lifting mechanism used for picking the pack up shall be capable of supporting the weight of the pack without breakage of the handle, its securing means, or that part of the product to which the handle is attached. Compliance is determined by		N/A

UL2271			
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	Section 35, Handle Loading Test.		
13	Electrical Spacings and Separation of Circuits		N/A
13.1	Electrical circuits within the EESA at opposite polarity shall be provided with reliable physical spacing to prevent inadvertent short circuits (i.e. electrical spacings on printed wiring boards, physical securing of uninsulated leads and parts, etc.). Insulation suitable for the anticipated temperatures and voltages shall be used where spacings cannot be controlled by reliable physical separation.		N/A
13.2	Electrical spacings in circuits shall have the following minimum over surface and through air spacings as outlined in Table 1 or the spacings requirements outlined in the Standard for Information Technology Equipment Safety – Part 1 : General Requirements, CAN/CSA-C22.2 No. 60950-1 / UL 60950-1 , the clause for "Clearances, Creepage Distances and Distances Through Insulation" .		N/A
13.3	There are no minimum spacings applicable to parts where insulating compound completely fills the casing of a compound or subassembly if the distance through the insulation, at voltages above 60 Vdc or above 30 Vrms is a minimum of 0.4-mm (0.02-in) thick for supplementary or reinforced insulation, and passes the Dielectric Voltage Withstand Test, Section 28 and the Isolation Resistance Test, Section 29. There is no minimum insulation thickness requirement for insulation of circuits at or below 60 Vdc or for basic or functional insulation. Some examples include potting, encapsulation, and vacuum impregnation.		N/A
13.4	Conductors of circuits operating at different voltages shall be reliably separated from each other through the use of mechanical securements such as barriers or wire ties to maintain spacings requirements unless they are each provided with insulation acceptable for the highest voltage involved. An insulated conductor shall be reliably retained so that it cannot contact an uninsulated live part of a circuit operating at a different voltage.		N/A
14	Insulation Levels and Protective Grounding		N/A
14.1	Hazardous voltage circuits shall be insulated from accessible conductive parts and safety extra low voltage (SELV) circuits as outlined in 14.2 through the following: a) Basic insulation and provided with a protective grounding system for protection in the event of a fault of the basic insulation; or b) A system of double or reinforced insulation; or c) A combination of (a) and (b).		N/A

UL2271			
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14.2	Safety extra low voltage (SELV) circuits (i.e. circuits at or below 60 Vdc or 48 Vrms under normal and single fault conditions) that are insulated from accessible conductive parts through functional insulation only are considered accessible.		N/A
14.3	If relying upon a protective grounding system (i.e. grounding of an accessible metal enclosure), it shall comply with 1 4.4 - 1 4.6.		N/A
14.4	Parts of a protective grounding system shall be reliably secured in accordance with 1 0.4 and provided with good metal-to-metal contact of the grounded parts of the EESA to the vehicle. The impedance from the various bonding conductors and connections to the main ground terminal shall have a maximum resistance of 0.1 Ω . Compliance can be determined by measurement using an ohmmeter.		N/A
14.5	The main ground terminal of the protective earthing ground system shall be identified by one of the following: a) A green-colored, not readily removable terminal screw with a hexagonal head; b) A green-colored, hexagonal, not readily removable terminal nut; c) A green colored pressure wire connector; or d) The word " Ground " or the letters " G " or " GR " or the grounding symbol (IEC 6041 7, No. 501 9 - upside down " tree " in circle) or otherwise identified by a distinctive green color.		N/A
14.6	Conductors, relied upon for the protective grounding and bonding system, shall be sized to handle intended fault current. If insulated, the insulation shall be green or green and yellow striped in color.		N/A
15	Protective Circuit and Safety Analysis		N/A
15.1	The EESA protective circuit shall maintain the cells within their normal operating region for charging and discharging through the life of the EESA. If normal limits are exceeded, the protective circuit shall limit or shut down the charging or discharging to prevent excursions beyond normal operating limits. Compliance is determined through a review of the pack and cell or electrochemical capacitor data including the safety analysis of 1 5.2 and through the testing of this standard.		N/A
15.2	An analysis of potential hazards (including a failure modes and effects analysis (FMEA)) shall be conducted on the EESA to determine that events that could lead to a hazardous condition have been identified and addressed through design or other means. Documents that can be used as guidance for the safety analysis include:		N/A

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	a) The Standard for Analysis Techniques for System Reliability - Procedure for Failure Mode and Effects Analysis (FMEA), IEC 6081 2; b) The Standard for Fault Tree Analysis (FTA), IEC 61 025; c) The Potential Failure Mode and Effects Analysis in Design (Design FMEA), Potential Failure Mode and Effects Analysis in Manufacturing and Assembly Processes (Process FMEA), SAE J1 739; d) The Procedures for Performing a Failure Mode, Effects, and Criticality Analysis, MIL-STD-1 629A; e) The Standard for Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems Part 1 : General Requirements, IEC 61 508-1 , and all parts.		
15.3	The analysis in 1 5.2 is utilized to identify anticipated faults in the system which could lead to a hazardous condition and the types and levels of protection provided to mitigate the anticipated faults. The analysis shall consider single fault conditions in the protection circuit/scheme as part of the anticipated faults.		N/A
15.4	When conducting the analysis of 1 5.2, active devices shall not be relied upon for critical safety unless: a) They are provided with a redundant passive protection device; or b) They are provided with redundant active protection that remains functional and energized upon loss of power/failure of the first level active protection; or c) They are determined to fail safe upon loss of power to the active circuit.		N/A
15.5	Devices relied upon for critical safety as noted in 1 5.4 shall be tested for functionality in accordance with appropriate functional safety requirements unless already evaluated through the other tests of this standard. Functional safety criteria for EESAs can be found in one of the following standard as appropriate to the design of the electronic and software protection scheme: a) The Standard for Safety Tests for Safety-Related Controls Employing Solid-State Devices, UL 991 and the Standard for Software in Programmable Components, UL 1 998; b) The Standard for Automatic Electrical Controls for Household and Similar Use - Part 1 : General Requirements, CSA-E60730-1 / UL 60730-1 ; c) The Standard for Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems Part 1 : General Requirements, IEC 61 508-1 , and all parts;		N/A

UL2271			
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	d) The Standard for Road Vehicles - Functional Safety - Part 1 : Vocabulary, ISO 26262-1 , and all parts.		
15.6	A hazardous voltage EESA shall have a manual disconnect to prevent inadvertent access to hazardous voltage parts during servicing or during a collision. The manual disconnect shall: a) Disconnect both poles of the hazardous voltage circuit; b) Be accessible and able to be operated without the use of a tool in the event of a collision or during servicing; c) Require manual action to break the electrical connection; d) Ensure disconnection is physically verifiable and can include actual removal of the EESA from the vehicle or unplugging the EESA connector/plug; and e) When engaged (i.e. under disconnection), it does not create exposed conductors capable of becoming energized and is insulated to prevent a shock hazard during actuation.		N/A
15.7	If a hazardous voltage automatic disconnect device is provided to isolate accessible conductive parts from the hazardous voltage circuit of the EESA, it shall: a) Not be able to be reset automatically although it may be able to be reset deliberately upon clearing of the fault; b) Disconnect both poles of the hazardous voltage circuit; c) Be capable of handling full load disconnects of the hazardous voltage circuit that it is isolating; and d) Not result in a hazardous condition upon automatic actuation.		N/A
16	Cells and Electrochemical Capacitors		P
16.1	Cells and electrochemical capacitors shall be designed to safely withstand anticipated abuse conditions for vehicular applications. Compliance is determined by the requirements in 1 6.2 - 1 6.6 and by the tests of this standard.		P
16.2	Lithium ion and other lithium based cells shall comply with the requirements for secondary lithium cells in the Standard for Lithium Batteries, UL 1 642.	Cells has been approved by UL 1642	P
16.3	Nickel metal hydride cells and other nickel based cells shall comply with the cell tests of the Testing Required for Cells table of the Standard for Household and Commercial Batteries, UL 2054.	Lithium ion battery	N/A
16.4	Sodium nickel metal chloride cells shall comply with the requirements for sodium beta battery cells outlined in Annex B of the Standard for Batteries For Use In Light	Lithium ion battery	N/A

UL2271			
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	Electric Rail (LER) Applications and Stationary Applications, UL 1 973.		
16.5	Lead Acid batteries shall comply with the pressure release test (if provided with a pressure release valve) and the flame arrester test (if provided with a flame arrester) of the Standard for Standby Batteries, UL 1 989.	Lithium ion battery	N/A
16.6	Electrochemical capacitors shall comply with the single capacitor requirements outlined in the Standard for Electrochemical Capacitors, UL 81 0A.		N/A
17	Manufacturing and Production Line Testing		P
17.1	EESAs shall be subjected to 1 00% production screening as described in 1 7.2 and 1 7.3 to determine the acceptability of spacing, insulation and grounding system in production.		N/A
17.2	A dielectric withstand test as outlined in the Dielectric Voltage Withstand Test in Section 29 shall be conducted on 1 00% production of EESAs with working voltage exceeding 60 Vdc or 30 Vrms/42.4 Vpeak.		N/A
17.3	A continuity check of the grounding conductors using an ohmmeter or other method shall be conducted on 1 00% production employing protective grounding. The continuity check shall determine that the resistance of the protective grounding system does not exceed 0.1 Ω .		N/A
PERFORMANCE			
18	General		P
18.1	Unless indicated otherwise, batteries shall be fully charged to the maximum operating state of charge (MOSOC) in accordance with the manufacturer's specifications for conducting the tests in this standard. After charging and prior to testing, the batteries shall be allowed to rest for a maximum period of 8 h at 25 $\pm$ 5° C (77 $\pm$ 9° F).	Tested as required.	P
18.2	Unless otherwise indicated, fresh samples representative of production are to be used for the tests described in this standard. The test program and number of samples to be used in each test is shown in Table 2.		N/A
18.3	All tests, unless noted otherwise, are conducted in a room ambient 25 $\pm$ 5° C (77 $\pm$ 9° F). Unless noted otherwise in the test method, the cooling systems or other auxiliary systems of the EESA may be allowed to operate during the test if they can affect the outcome of the test.	See tests below.	P
18.4	Temperature shall be measured using thermocouples consisting of wires not larger than 0.21 mm 2 (24 AWG) and not smaller than 0.05 mm 2 (30 AWG)		P

UL2271			
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	connected to a potentiometer-type instrument. Temperature measurements are to be made with the measuring junction of the thermocouple held tightly against the component/location being measured. For those tests that require the sample to reach thermal equilibrium (also referred to as steady state conditions), thermal equilibrium is considered to be achieved if after three consecutive temperature measurements taken at intervals of 1 0% of the previously elapsed duration of the test but not less than 1 5 min, indicate no change in temperature greater than $\pm 2^{\circ}\text{C}$ ($\pm 3.6^{\circ}\text{F}$).		
18.5	Where there is a specific reference to a single fault condition in the individual test methods, the single fault is to consist of a single failure (i.e. open, short or other failure means) of any component in the EESAs that could occur and affect the results of the test. This fault is implemented in conjunction with the test being conducted (i.e. overcharge, short circuit, etc.) or may be conducted as part of a verification of a protective circuit. A protective device determined to be reliable may remain in the circuit without being faulted. See Annex A and 2.1 . A protective device determined to be reliable is one that has been shown to comply with an appropriate component safety standard and is used within its ratings.		P
18.6	The tests contained in this standard may result in explosions, fire and emissions of flammable and/or toxic fumes as well as electric shock. It is important that personnel use extreme caution and follow local and regional worker safety regulations when conducting any of these tests and that they be protected from flying fragments, explosive force, and sudden release of heat and noise that could result from testing. The test area is to be well ventilated to protect personnel from possible harmful fumes or gases. As an additional precaution, the temperatures on surface of at least one cell/module within the DUT and the OCV of the DUT are to be monitored during the test for safety and information purposes. All personnel involved in the testing are to be instructed to never approach the DUT until temperatures are falling and have returned to within ambient temperatures		P
18.7	Unless noted otherwise in the individual test methods, the tests shall be followed by a minimum 1 -h observation time prior to concluding the test and temperatures are to be monitored in accordance with 1 8.6.		
19	Combustible Concentrations		N/A

UL2271			
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19.1	Where a system analysis indicates that venting/off gassing from cells or capacitors could result in combustible concentrations because of the system' s chemistry (i.e. vented batteries) or design, the evaluation for combustible concentrations during the tests of this standard shall be determined through the methods outlined in 1 9.2. If it can be determined through examination of the cells after testing that they did not vent as a result of the test, the system is in compliance with these criteria.		N/A
19.2	For detection of potential flammable concentrations that may be emitted during testing, a gas monitor suitable for detecting 25% of the lower flammability limit of the evolved gases being measured. A minimum of two sampling locations where concentrations may occur such as at vent openings or vent ducts shall be used for taking measurements.		N/A
19.3	Additional precautions shall be taken during tests requiring this analysis due to the potential for flammable gas concentrations that may occur within the test room or chamber.		N/A
20	Tolerances		P
20.1	Unless noted otherwise in the test methods, the overall accuracy of controlled or measured values when conducting testing in accordance with this standard shall be within following tolerances: a) ± 1 % for voltage; b) ± 1 % for current; c) $\pm 2^{\circ}$ C ($\pm 3.6^{\circ}$ F) for temperatures at or below 200° C (392° F) and $\pm 3\%$ for temperatures above 200° C (392° F); d) ± 0.1 % for time; and e) ± 1 % for dimension.	Test facilities checked and complied.	P
20.2	These tolerances comprise the combined accuracy of the measuring instruments, the measurement techniques used, and all other sources of error in the test procedure.	Temperature on battery case recorded for so requested clauses, also see appended tables.	P
21	Post Test Cycle		P
21.1	EESAs that are operational after the following tests shall be subjected to a minimum of one cycle of charging and discharging in accordance with the manufacturer' s specifications to determine that there is no non-compliant results as outlined in Table 3 for that test: a) Electrical Tests - Overcharge, short circuit, overdischarge protection, imbalanced charging; b) Mechanical Tests - Vibration, shock, roll over, drop;		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	c) Environmental Tests - Water exposure, and thermal cycling.		
22	Results Criteria		P
22.1	See Table 3 for results criteria for tests outlined in this standard and Section 6 for definitions of the non-compliance results terms. See also individual tests methods for any additional details.		P
ELECTRICAL TESTS			
23	Overcharge Test		P
23.1	<i>This test is intended to evaluate an DUT' s ability to withstand an overcharge condition under a single fault in the charging control circuitry that could result in an overcharge condition.</i>		P
23.2	A fully charged sample is to be discharged at a 0.2 C constant discharge rate or a higher discharge rate permitted by the manufacturer to the manufacturer' s specified EODV. The DUT is then subjected to a constant current charging at the manufacturer' s maximum specified charging rate and under a single fault condition in the circuitry that directly controls the charging line of the DUT that could lead to an overcharge condition. Protective devices that have been determined reliable may remain in the circuit as noted in 1 8.5. For information purposes, temperatures are to be monitored on the cell/module where temperatures may be highest.		P
23.3	The test is to be continued until the voltage has reached 1 1 0% of the maximum specified voltage limit and monitored temperatures return to ambient or steady state conditions and an additional 2 h has elapsed, or explosion/fire occur. If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
23.4	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P
23.5	If a protective device in the circuit operates, the test is repeated at 90% of the trip point of the protection device or at some percentage of the trip point that allows charging for at least 1 0 min. Temperatures shall be measured on the DUT for monitoring purposes.		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
23.6	As a result of the overcharge test, any of the following results in (a) - (g) below are considered a non-compliant results. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		P
24	Short Circuit Test		P
24.1	This test evaluates a DUT' s ability to withstand a short circuit condition.	Tested complied.	P
24.2	A fully charged sample of the DUT is to be short-circuited by connecting the positive and negative terminals of the sample with a circuit load having a total resistance of less than or equal to 20 mΩ ..	Tested complied.	P
24.3	Samples are to be subjected to a single fault across any protective device in the load circuit. Protective devices that have been determined reliable may remain in the circuit as noted in 1 8.5.		P
24.4	The sample shall be discharged until the sample has returned to ambient temperature or fire or explosion occurs. Temperatures shall be measured on the DUT for monitoring purposes.		P
24.5	If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
24.6	If a protective device in the circuit operates, the test is repeated at 90% of the trip point of the protection device or at some percentage of the trip point that allows discharging for at least 1 0 min.		P
24.7	At the conclusion of the test and after cooling to near ambient, the samples that contain hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P
24.9	As a result of the short circuit test, any of the following results in (a) - (g) below are considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire;		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		
25	Overdischarge Test		P
25.1	Primary batteries (for example: cells, single cell batteries, or battery packs) shall comply with 10.2 –10.5(Secondary battery pack)		P
25.2	The fully charged sample is to be subjected to a constant discharging current at the maximum discharging current specified by the manufacture under a single fault condition in the discharging circuit of the DUT that could lead to an overdischarge condition. Protective devices that have been determined reliable may remain in the circuit as noted in 1 8.5. Temperatures shall be measured on a cell/module for monitoring purposes.		P
25.3	The test is to be continued until the sample is fully discharged to a near zero voltage state or protective devices remaining in the circuit operate, and the monitored temperatures return to ambient or steady state, or explosion and/or fire occurs. If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
25.4	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to an Isolation Resistance Test (without humidity conditioning) or a Dielectric Voltage Withstand Test.		P
25.5	As a result of the overdischarge test, any of the following results in (a) - (g) below are considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		P
26	Temperature Test		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
26.1	This test is conducted to determine whether or not the modules and their cells are being maintained within their specified operating limits during maximum charge and discharge conditions of the EESA. During this test, it shall also be determined as to whether or not temperature sensitive safety critical components and temperature sensitive materials in the DUT are being maintained within their temperature ratings based upon the maximum operating temperature limits of the DUT. Temperatures on accessible surfaces, which may be contacted by the user, are also monitored.		P
26.2	A fully discharged DUT (i.e. discharged to EODV) is to be conditioned within a chamber set to the upper limit charging temperature specifications of the EESA. After thermal stabilization in the chamber, the DUT is to be connected to a charging circuit input representative of anticipated maximum charging parameters. The DUT shall then be subjected to maximum normal charging while monitoring voltages and currents on cells/modules until it reaches the manufacturer' s specified fully charged condition. Temperatures shall be monitored on temperature sensitive components including cells and on any user accessible surfaces.		P
26.3	While still in the conditioning chamber, and after allowing temperatures to stabilize, the fully charged DUT shall then be discharged in accordance with the manufacturer' s specifications down to the manufacturer' s specified end of discharge condition while monitoring voltage and current on cells/modules until the DUT reaches its specified EODV. Temperatures shall be monitored on temperature sensitive safety critical components including cells and on any user accessible surfaces.		P
26.4	The charge and discharge cycles are then repeated for a total of 5 complete cycles of charge and discharge.		P
26.5	During the temperature test, the voltage and current during discharge and charging of the component cells is monitored to determine that they are not outside of the specified cell manufacturer's operating region.		P
26.6	The manufacturer' s specified limits (voltage, current and temperatures measured) shall not be exceeded during the charging and discharging cycles. Temperatures measured on components shall not exceed their specifications. See Tables 4 and 5 for surface and component temperature limits.		P
26.7	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to an Isolation Resistance Test (without humidity conditioning)		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	or a Dielectric Voltage Withstand Test.		
26.8	As a result of the temperature test, any of the following results in (a) - (h) below are also considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) V - Venting; g) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); h) P - Loss of protection controls.		P
27	Imbalanced Charging Test		P
27.1	This test is to determine whether or not an DUT with series connected cells can maintain the cells within their specified operating parameters if it becomes imbalanced.	Tested and passed	P
27.2	A fully charged DUT shall have all of its cells with the exception of one cell/cell block discharged to its specified fully discharged condition. The undischarged cells shall be discharged to approximately 50% of its specified state of charge (SOC) to create an imbalanced condition prior to charging.		P
27.3	The DUT shall then be charged in accordance with the manufacturer' s specifications. The voltage of the partially charged cells shall be monitored during the charging to determine if its voltage limits are exceeded. If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values.		P
27.4	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to an Isolation Resistance Test (without humidity conditioning) or a Dielectric Voltage Withstand Test.		P
27.5	The maximum voltage limit of the cells shall not exceed the manufacturer' s specifications. In addition, any of the following results in (a) - (h) below are considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure);		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	e) L - Electrolyte Leakage (external to enclosure); f) V - Venting; g) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); h) P - Loss of protection controls.		
28	Dielectric Voltage Withstand Test		N/A
28.1	This test is an evaluation of the electrical spacings and insulation at hazardous voltage circuits within the DUT.		N/A
28.2	Circuits at 60 Vdc or higher shall be subjected to a dielectric withstand voltage consisting of a dc potential of twice the rated voltage times 1.41 4.	<60V	N/A
28.3	The test voltage is to be applied between the hazardous voltage circuits of the DUT and non-current carrying conductive parts that may be accessible.		N/A
28.4	The test voltage is also to be applied between the hazardous voltage charging circuit and the enclosure/accessible non-current carrying conductive parts of the DUT.		N/A
28.5	If the accessible parts of the DUT are covered with insulating material that may become live in the event of an insulation fault, then the test voltages are applied between each of the live parts and metal foil in contact with the accessible parts. The metal foil shall be wrapped tightly around and in intimate contact with the accessible part. The foil is to be drawn tightly across any opening in the enclosure or other accessible parts to form a flat plane across such opening. See Figure 3.		N/A
28.6	The test voltages shall be applied for a minimum of 1 min with the cells/modules disconnected to prevent charging during application of the voltage.		N/A
28.7	The test equipment shall consist of a 500 VA or larger capacity transformer, the output voltage, which is variable and which is essentially sinusoidal if using ac test method a dc output if using a dc test method. There is no trip current setting for the test equipment since the test is checking for insulation breakdown, which results in a large increase of current. Setting a trip current may result in a false failure of this test, as it may not be indicative of insulation breakdown.		N/A
28.8	There shall be no evidence of a dielectric breakdown (breakdown of insulation resulting in a short through insulation/arcing over electrical spacings) as evidenced by an appropriate signal from the dielectric withstand test equipment as a result of the applied test voltage. Corona discharge or a single momentary discharge is not regarded as an dielectric breakdown (i.e. insulation breakdown).		N/A
29	Isolation Resistance Test		N/A

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
29.1	Isolation resistance method for systems rated 120 V and above		N/A
29.1.1	This test is intended to determine that insulation provides adequate isolation of hazardous voltage circuits from accessible conductive parts of the DUT and that the insulation is non-hygroscopic.		N/A
29.1.2	The electric energy storage assembly shall be subjected to Isolation resistance test of Clause 6.1 .3 of the Standard for Electrically Propelled Road Vehicles - Safety Specifications - Part 1 : On-board Rechargeable Energy Storage System (RESS), ISO 6469-1 , with the following exceptions in (a) and (b). The DUT is to be in the fully charged state for both (a) and (b): a) Testing is to be conducted with the DUT in the as-received condition (i.e. no humidity conditioning prior to measurements); and b) Testing is to be conducted after conditioning the DUT in accordance with the Standard for Environmental Testing - Part 2-30: Tests - Test Db: Damp Heat, Cyclic (1 2 h + 1 2 h Cycle), IEC 60068-2-30 using the following parameters: 1) Variant 1 ; 2) At maximum temperature of $55 \pm 2^{\circ} \text{C}$ ($131 \pm 3^{\circ} \text{F}$); and 3) 6 cycles.		N/A
29.1.3	For condition 29.1 .2(b), the DUT shall be placed within the chamber so that it is oriented in the manner in which it will be installed in the vehicle. Upon completion of the 6th cycle the sample shall be subjected to a controlled recovery in accordance with Recovery, Clause 9, of the Standard for Environmental Testing - Part 2-30: Tests - Test Db: Damp Heat, Cyclic (1 2 h + 1 2 h Cycle), IEC 60068-2-30. The isolation resistance measurements shall be made within 30 min of completion of the controlled recovery phase.		N/A
29.1.4	For both (a) and (b) of 29.1 .2, the isolation resistance divided by the maximum working voltage of the circuit under test, shall be at least $100 \Omega / \text{V}$ for dc systems and $500 \Omega / \text{V}$ for ac systems or combined ac and dc systems.		N/A
29.2	Isolation resistance method for systems rated below 120 V (insulation resistance method)		N/A
29.2.1	This test is intended to determine that insulation of the DUT provides adequate isolation of hazardous voltage circuits from accessible conductive parts of the DUT and that the insulation is non-hygroscopic.		N/A

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
29.2.2	A DUT with accessible parts shall be subjected to an insulation resistance test between the positive terminal and accessible dead metal parts of a DUT. If the accessible parts of the DUT are covered with insulating material that may become live in the event of an insulation fault, then the test voltages are applied between each of the live parts and metal foil in contact with the accessible parts as shown in 28.5 and Figure 3.		N/A
29.2.3	The insulation resistance shall be measured after a 60-s application with a high resistance voltmeter using a 500 Vdc potential applied for at least 1 min to the locations under test.		N/A
29.2.4	The test shall be repeated on a sample subjected to humidity conditioning in accordance with the Standard for Information Technology Equipment Safety – Part 1 : General Requirements, CAN/CSA-C22.2 No. 60950-1 / UL 60950-1 , Clause 2.9.2. Measurements shall be made with the sample still in the chamber.		N/A
29.2.5	The measured insulation resistance between the positive terminals and accessible parts of the DUT shall be at least 50,000 Ω .		N/A
30	Vibration Endurance Test		P
30.1	This test evaluates the DUT' sability to withstand vibration that may occur from its use in an LEV. The test shall be performed in accordance with the Standard for Electrically Propelled Road Vehicles – Test Specification for Lithium-Ion Traction Battery Packs and Systems – Part 1 : High-Power Applications, ISO 1 2405-1 , (which references the Standard for Environmental Testing – Part 2-64: Tests – Test Fh: Vibration, Broadband Random Guidance, IEC 60068-2-64, Tables 20 to 23), or according to a test profile determined by the customer and verified to the LEV application.	Tested and passed	P
30.2	The DUT is to be securely mounted to a vibration test platform in a manner similar to how it is mounted in the end use LEV located within a chamber or test room, where the temperature during testing can be varied. The DUT is to be subjected to a random vibration along three perpendicular axes in space in a sequence starting with the vertical axes (Z) and ending with the longitudinal axis (X).		P
30.3	The DUT shall be subjected to the vibration in each axis for 21 h if testing one sample, 1 5 h is testing two samples or 1 2 h if testing 3 samples. For each axis the frequency shall be varied from 5 Hz to 200 Hz with power spectral density (PSD) as outlined in the Table 6.		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
30.4	While subject to vibration, the DUT shall also be subject to variations in temperature during vibration as outlined in Table 7. Monitoring for information purposes per 1 8.6 should be conducted during the test.		P
30.5	If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
30.6	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P
30.7	As a result of the vibration endurance test, any of the following results in (a) - (g) below are considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		P
31	Shock Test		P
31.1	This test is intended to determine whether or not the DUT can withstand a mechanical shock that may occur when in use in an LEV.	Tested and passed	P
31.2	The fully charged sample is to be secured to the testing machine by means of a rigid mount, which supports all mounting surfaces of the sample. Temperatures on the center cell are monitored for information purposes.		P
31.3	The sample is to be subjected to mechanical shock testing with parameters as shown in Table 8 or according to a test profile determined by the customer and verified to the LEV application. The shocks are to be applied in all 6 spatial directions for prismatic DUT and 3 spatial directions for cylindrical DUT.		P
31.4	During the test the OCV of the sample and temperatures on the center cell(s/modules) may be monitored for information purposes.		P
31.5	If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values. The test shall be followed by an observation period per 1 8.7.		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
31.6	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P
31.7	As a result of the shock test, any of the following results in (a) - (g) below are considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		P
32	Crush Test		P
32.1	This test is conducted to determine the DUT' s ability to withstand a crush that could occur during a vehicle accident. This test is only applicable to on road LEVs such as scooters and motorcycles that could be involved in a vehicle crash.	Tested and passed	P
32.2	This test is conducted on a fully charged DUT.		P
32.3	A sample shall be crushed between a fixed surface and a ribbed test platen. The platen shall have semi-cylindrical intruders that have a 75-mm (2.95-in) radius and placed 30-mm (1.2-in) from one another across the face of the platen as shown in Figure 4.		P
32.4	The sample is to be subjected to a crushing force that deforms the DUT by 50% but no greater force than 1 ,000 times weight of the DUT or 1 00 kN, whichever comes first. The crushing force is to be applied at a rate of approximately 1 .5 cm/s upon first contact with the sample. The crush is to be applied in each of 3 mutually perpendicular directions with the ribs centered on the sample in each direction of press.		P
32.5	Each sample shall only subject to one crush. The force shall be increase from zero to a maximum force value (i.e. 1 ,000 times the weight of the DUT or 1 00 kN) or when the maximum of 50% deformation is achieved, whichever comes first and then it is removed. Monitoring for information purposes per 1 8.6 should be conducted during the test.		P
32.6	At the conclusion of the observation period, the samples are examined for electrolyte leakage and samples with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
32.7	As a result of the crush test, any of the following results in (a) and (e) below are considered non-compliant results. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) L - Electrolyte Leakage (external to enclosure); e) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown);		P
33	Drop Test		P
33.1	Drop test for EESAs intended to be removable by the user	Tested and passed	P
33.1.1	This test is intended to evaluate whether a hazard exists when an DUT is subjected to an inadvertent drop during lifting or handling by the user when charging or replacement, etc.		P
33.1.2	A fully charged DUT is to be dropped three times from a height of 1.0 ± 0.01 m (39.4 ± 0.4 in) to strike a flat concrete surface in a manner most representative of what would occur during lifting or handling of the DUT by the user. The concrete surface shall be at least 76-mm (3-in) thick and shall be large enough in area to cover the DUT. If the DUT is operational after the drop, it is to be subject to a minimum of one normal charge/discharge cycle in accordance with the manufacturer's specifications. The DUT shall be subjected to an observation period per 1 8.7 and then examined. Monitoring for information purposes per 1 8.6 should be conducted during the test.		P
33.1.3	If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer's maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
33.1.4	After the examination, the DUTs shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning) if applicable.		P
33.1.5	There shall be no damage of the enclosure that would allow hazardous voltage parts to be accessed by use of the articulate test finger shown in Figure 1 and test pin shown in Figure 2.		P
33.1.6	As a result of the drop test, any of the following results in (a) - (g) below are considered non-compliant. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		
33.2	Drop test for EESAs intended for service handling only		N/A
33.2.1	This test is intended to evaluate whether a hazard exists when a DUT is subjected to an inadvertent drop during installation or removal from the vehicle.		N/A
33.2.2	After being equilibrated at room temperature, a fully charged DUT is to be dropped from a minimum height of 1.0 m (3.3 ft) to strike a flat concrete surface in the position most likely to produce the adverse results and in a manner and height most representative of what would occur during maintenance and handling/removal of the DUT during servicing.		N/A
33.2.3	If only one drop test is performed, it shall not be a flat drop, and if one drop test is a flat drop, then at least one other test shall be performed that is not a flat drop.		N/A
33.2.4	The DUT is to be dropped a minimum of one time.		N/A
33.2.5	The concrete surface is to be at least 76-mm (3-in) thick and shall be large enough in area to cover the DUT.		N/A
33.2.6	The DUT shall be examined within a time frame of 6 - 24 h after dropping. During the test, temperatures shall be monitored. If the DUT returns to or remains near ambient, a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning) is to be conducted.		N/A
33.2.7	As a result of the drop test, any of the following results in (a) - (e) are considered non-compliant. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) L - Electrolyte Leakage (external to enclosure); e) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown);		N/A
34	Mold Stress Relief Test		N/A
34.1	This test is intended to evaluate whether any shrinkage or distortion exist on a molded or formed thermoplastic enclosure due to release of internal stresses caused by the molding or forming operation and result in the exposure of hazardous parts or reduction of electrical spacings.		N/A
34.2	The sample is to be placed in a full-draft circulating-air		N/A

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	oven maintained at a uniform temperature of 70° C (158° F). The samples are to remain in the oven for 7 h.		
34.3	To prevent hazards from overheating energized cells, samples shall be fully discharged prior to conditioning.		N/A
34.4	After careful removal from the oven, the sample shall be allowed to cool to room temperature and then examined. After the examination, the samples shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		N/A
34.5	There shall be no insulation breakdown during the Dielectric Voltage Withstand Test or the isolation resistance shall not be below the levels outlined in Section 29.		N/A
34.6	There shall be no damage of the DUT enclosure that would allow hazardous voltage parts to be accessed by use of the articulate test finger shown in Figure 1 and the test pin shown in Figure 2.		N/A
35	Handle Loading Test		N/A
35.1	This test is intended to evaluate the strength of the handle(s) on a user removable EESA.		N/A
35.2	A force is to be applied on the handle in the intended carrying direction uniformly over a 75-mm (2.95-in) length at the center of the handle. The applied force shall be gradually increased from zero to four times the weight of the DUT in 5 - 10 s and then maintained at the level for 1 min.		N/A
35.3	If more than one handle is provided, the test force shall be determined by the percentage of the DUT weight sustained by each handle with the DUT in the intended carrying position. If a DUT weighing less than 25 kg (55.1 lbs) is provided with more than one handle and can be carried by only one handle, each handle shall be capable of withstanding a force based on the total weight of the DUT.		N/A
35.4	There shall be no breakage of the handle, its securing means, or that part of the DUT to which the handle is attached.		N/A
36	Roll Over Test		P
36.1	This test is to evaluate whether or not a rotation of an EESA as might occur within an LEV that is turned over could lead to a hazardous condition. A sample of the electric energy storage assembly shall be subjected to a 360° rotation to simulate an overturned vehicle.	Tested and passed	P
36.2	The fully charged sample shall be rotated at a continuous rate of 90° /15 s. Testing shall be conducted subjecting the sample to a 360° rotation in 3 mutually perpendicular different directions. DUTs with only 2 axes of symmetry, such as cylindrical		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	designs are subjected to 2 mutually perpendicular directions of rotation. Monitoring for information purposes per 1 8.6 should be conducted during the test.		
36.3	If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
36.4	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P
36.5	As a result of the roll over test, any of the following results in (a) - (g) below are considered non-compliant. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		P
37	Strain Relief Tests (Cord Anchorages)		N/A
37.1	General		N/A
37.1.1	The strain relief tests are conducted on those EESAs that have exposed non-detachable cords or cables that may be subjected to pull in the end use vehicle.		N/A
37.2	Strain relief pull test		N/A
37.2.1	The purpose of this test is to determine if the strain relief means for a non-detachable accessible cord prevents damage or displacement upon being pulled.		N/A
37.2.2	One sample of the EESA or accessory provided with a strain relief shall withstand without damage to the cord or conductors and without displacement, a direct pull of 2 times the weight of the DUT but no greater than 1 56 N (35 lbf), applied to the cord for 1 min. Supply connections within the equipment are to be disconnected from terminals or splices during the test when applicable.		N/A
37.2.3	If the cord anchorage is mounted in polymeric enclosure material, the test is to be conducted after the mold stress test and after the sample has cooled to room temperature.		N/A
37.2.4	As a result of the pull force, there was no damage or displacement of internal connectors. Inner conductors may not elongate more than 2 mm (0.08 in) from the		N/A

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	pre-test position.		
37.3	Push-back test		N/A
37.3.1	The purpose of this test is to determine if the strain relief of a non-detachable accessible cord provides adequate protection to connections and prevents hazardous displacement of internal wiring and connections as a result of push back.		N/A
37.3.2	The DUT is to be tested in accordance with 37.3.3 and 37.3.4 without occurrence of any of the following conditions: a) Subjecting the supply cord to mechanical damage; b) Exposing the supply cord to a temperature higher than that for which it is rated; c) Reducing spacings (such as to a metal strain-relief clamp) below the minimum required values; or d) Damaging internal connections or components.		N/A
37.3.3	The non-detachable cord is to be held 25.4 mm (1 in) from the point where it emerges from the DUT and is then to be pushed back into the DUT. When a removable bushing, which extends further than 25.4 mm (1 in) is present it is to be removed prior to the test.		N/A
37.3.4	When the bushing is an integral part of the cord, then the test is to be carried out by holding the bushing. The cord is to be pushed back into the product in 25.4-mm (1-in) increments until the cord buckles or the force to push the cord into the product exceeds 26.7 N (6 lbf).		N/A
37.3.5	The cord is to be manipulated to determine compliance with 37.3.2.		N/A
ENVIRONMENTAL TESTS			
38	Immersion Test		P
	This test is intended for those electric energy storage assemblies that may be subjected to potential immersion in the LEV application. (i.e. underbody location). This test is conducted on a fully charged (i.e. MOSOC)DUT and is intended to determine its resistance to a hazardous condition as a result of a temporary immersion in water from flooding of a vehicle.		P
	While the DUT is in its normal operating orientation and with switches/contactors in closed position, it shall be subjected to an immersion. The DUT is to be immersed in salt water (5% by weight NaCl in H ₂ O) at room temperature for a minimum of 2 h or until any visible reactions have stopped. The water depth is to be sufficient to completely submerge the DUT.		P
	After removal from the water, the DUTs with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	As a result of the water immersion test, any of the following results in (a) - (c) below are considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown);		P
39	Water Exposure Test (IP Code Rating)		P
	This test is intended to evaluate the EESA' s ability to withstand potential water exposure in its intended use such as cleaning, outdoor storage or use, etc., and is conducted in accordance with the test method outlined in 39.2.		P
	A fully charged DUT (i.e. MOSOC) shall be subjected to a water exposure test in accordance with the Standard for Degrees of Protection Provided by Enclosures (IP Code), IEC 60529, Tests for protection against water indicated by the second characteristic numeral 3 (IPX3) unless the EESA is provided with a higher IP Code rating, in which case the DUT shall be tested in accordance with its rating. Monitoring for information purposes per 1 8.6 should be conducted during the test.	IP64	P
	If the DUT is operational after the test, it shall be subjected to a minimum of one charge/discharge cycle at the manufacturer' s maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P
	As a result of the water exposure test, any of the following results in (a) - (g) below are considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		P
40	Thermal Cycling Test		P
	This test determines the electrical EESA' s ability to withstand exposure to rapidly changing environments such as when the vehicle is entering or exiting a heated garage after being in a cold environment, or during		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	transport etc. without evidence of damage that could lead to a hazardous event.		
	A fully charged DUT (i.e. MOSOC) shall be subjected to the thermal shock test of the Standard for Electric and Hybrid Electric Vehicle Rechargeable Energy Storage System (RESS) Safety and Abuse Testing, SAE J2464, except that the temperature extremes are from $85 \pm 2^{\circ}\text{C}$ to $-40 \pm 2^{\circ}\text{C}$ ($185 \pm 3.6^{\circ}\text{F}$ to $-40 \pm 3.6^{\circ}\text{F}$). Monitoring for information purposes per 1 8.6 should be conducted during the test.		P
	If the DUT is operational after the test, it shall be allowed to return to room ambient and then subjected to a minimum of one charge/discharge cycle at the manufacturer's maximum specified values. The test shall be followed by an observation period per 1 8.7.		P
	At the conclusion of the observation period, the samples with hazardous voltage circuits shall be subjected to a Dielectric Voltage Withstand Test or Isolation Resistance Test (without humidity conditioning).		P
	As a result of the thermal cycling test, any of the following results in (a) - (g) below considered a non-compliant result. See also Table 3 and Section 22. a) E - Explosion; b) F - Fire; c) C - Combustible Concentrations (if applicable to technology); d) R - Rupture (enclosure); e) L - Electrolyte Leakage (external to enclosure); f) S - Electric shock hazard (resistance below isolation resistance limits or dielectric breakdown); g) P - Loss of protection controls.		P
41	Label Permanence Test		P
41.1	The purpose of this test is to evaluate the permanence of an adhesive label that has not been subjected to a previous evaluation program.		P
41.2	An adhesive label secured to a surface representative of the end use application and is subjected to the following conditioning: a) The label sample is rubbed by hand for 15 s with a piece of cloth soaked with water; and b) The sample is again rubbed for 15 s with a piece of cloth soaked with petroleum spirit.		P
	The petroleum spirit to be used for the test is an aliphatic solvent hexane having: a) A maximum aromatics content of 0.1 % by volume; b) A kauributenol value of 29; c) An initial boiling point of approximately 65°C (149°F).		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	F); d) A dry point of approximately 69° C (1 56.2° F); and e) A mass per unit volume of approximately 0.7 kg/l.		
	After the conditioning outlined in 41 .2, the sample is to be examined for signs of damage including curing and to determine if the marking is still legible. The sample is also examined to determine if it can be removed by easily by hand from the surface the adhered surface.		P
	As a result of the conditioning, the sample label shall remain legible, show no evidence of damage including curling and shall not be able to be able to be easily removed by hand from the adhered surface.		P
MARKINGS			
42	General		P
42.1	The markings required for compliance to this standard shall be legible and permanent such as etched, adhesive labels, etc. An adhesive-backed label shall comply with the requirements in the Standard for Marking and Labeling Systems, UL 969, for the intended exposure conditions and surface adhered to.		P
42.2	EESAs are to be marked with the manufacturer' s name, trade name, trademark or other descriptive marking which may identify the organization responsible for the product, part number or model number, and electrical ratings in volts dc and Ah or Wh, and chemistry.		P
42.3	Terminals shall be marked " positive " or " + " and " negative " or " - " or both to indicate polarity unless the terminal is keyed in a manner that prevents incorrect connections in the end use vehicle.		P
42.4	EESAs shall also be marked with the date of manufacture, which may be in the form of a code that does not repeat within 1 0 years.		P
42.5	EESAs shall be marked with charging instructions. An example of such markings would be the following or equivalent " Use Only () Charger " .		P
42.6	All external terminals and connections shall be provided with identification and if applicable, polarity markings.		P
42.7	The point of connection to the charger earth grounding system shall be identified by the word " Ground " or the letters " G " or " GR " or the grounding symbol IEC 6041 7, No. 501 9 (upside down tree within a circle) or otherwise identified by a distinctive green color. Any other grounding terminals shall also be identified in a manner that is distinctive from the main earth ground terminal for the charger system.		P
42.8	The EESA that contains hazardous voltage circuits shall		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	be marked with the following or be marked with the electric shock hazard symbol ISO 3864 No. 5036 (lightning bolt within a triangle):		
	In Canada: In English: " Warning: Hazardous Voltage Circuits " ; and In French: «Attention: Circuits à tension élevée» In the United States: In English: " Warning: Hazardous Voltage Circuits " .		N/A
42.9	EESAs shall be marked as follows or with the sign M002 of the Standard for Graphical Symbols Safety Colours and Safety Signs, Safety Signs Used in Workplaces and Public Area, ISO 701 0, No. W001 (i.e. exclamation point in triangle): In Canada: In English: " WARNING - To reduce the risk of injury, user must read instruction manual " ; and In French: «AVERTISSEMENT - Pour prévenir les blessures, l' utilisateur doit lire le manuel d' utilisation» In the United States: In English: " WARNING - To reduce the risk of injury, user must read instruction manual " .		N/A
42.10	An EESA may or may not be marked with the minimum required IPX3 rating. EESAs marked with a higher IP rating than the minimally required rating of IPX3, shall comply with the requirements for that higher rating in accordance with Section 39.		N/A
INSTRUCTIONS			
43	General		P
43.1	An EESA shall be provided with complete instructions for installation and integration in the end use vehicle including instructions for how to secure the EESA in the end use vehicle to prevent movement and stress on connections and parts.		P
43.2	An EESA shall be provided with instructions for the proper use including charging and discharging, storage and disposal. These instructions shall include temperature limits, charging and discharging limits. Instructions for replacement of user replaceable fuses shall also be provided.		P
43.3	An EESA intended for removal and charging outside of the vehicle shall be provided with instructions for the safe handling including removal and insertion into the vehicle and during charging and instructions for storage of fully charged EESA.		P
43.4	The following or equivalent marking shall be provided in the instructions:		P

UL2271			
Clause	Requirement + Test	Result - Remark	Verdict
	In Canada: In English: " WARNING - Risk of Fire and Electric Shock - No User Serviceable Parts " ; and In French: «AVERTISSEMENT - Risque d' incendie et de choc électrique - Aucune des pièces ne peut être réparée par l' utilisateur» In the United States: In English: " WARNING - Risk of Fire and Electric Shock - No User Serviceable Parts " . In Canada: In English: " WARNING - Risk of Fire - No User Serviceable Parts " ; and In French: «AVERTISSEMENT - Risque d' incendie - Aucune des pièces ne peut être réparée par l' utilisateur» In the United States:		

23	TABLE: Overcharge Test					P
Ambient temperature: 24.5°C						
Sample No.	01#	02#	03#	04#	05#	
Charge Current	27.5A	27.5A	27.5A	27.5A	27.5A	
Cell case temp (°C)	83.5	84.6	84.3	83.9	84.2	
Battery case temp (°C)	81.4	82.3	82.0	81.8	81.6	
Failure Mode	Yes	Yes	Yes	Yes	Yes	
Faulted Protective Device	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	
Supplementary information: No explosion or fire; Tmax was recorded on the centre of the cell/battery surface. Faulted Protective Device: MOSFET (Q1) is bypassed during the test. A PTC limits the input current value during the test. S-C: Short circuit.						

24	TABLE: Short-Circuit Test					P
Ambient temperature: 24.4°C						
Sample No.	06#	07#	08#	09#	10#	
Cell case temp (°C)	76.3	76.6	75.2	76.9	75.5	
Battery case temp (°C)	73.2	73.9	72.6	73.1	72.0	
Failure Mode	Yes	Yes	Yes	Yes	Yes	
Faulted Protective Device	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	--	
Ambient temperature: 60.0°C						
Sample No.	06#	07#	08#	09#	10#	
Cell case temp (°C)	97.6	98.3	98.7	98.2	99.3	
Battery case temp (°C)	93.5	93.1	94.1	94.6	93.7	
Failure Mode	Yes	Yes	Yes	Yes	Yes	
Faulted Protective Device	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	
Supplementary information: No explosion or fire, or chemical leak; Tmax was recorded on the centre of the cell/battery surface. Faulted Protective Device: MOSFET (Q1) is bypassed during the test. A PTC limits the input current value during the test. S-C: Short circuit.						

25	TABLE: Overdischarge Test					P
Ambient temperature: 24.5°C						
Sample No.	11#	12#	13#	14#	15#	

discharge Current	55A	55A	55A	55A	55A
Cell case temp (°C)	93.4	92.7	93.5	94.6	95.1
Battery case temp (°C)	89.2	88.3	90.0	89.4	88.8
Failure Mode	Yes	Yes	Yes	Yes	Yes
Faulted Protective Device	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C	MOSFET (Q1) S-C
Supplementary information: No explosion or fire; Tmax was recorded on the centre of the cell/battery surface. Faulted Protective Device: MOSFET (Q1) is bypassed during the test. A PTC limits the input current value during the test. S-C: Short circuit.					

26	Temperature Test					P
Ambient temperature: see below						
Sample No.	16#		17#		Limited T	
Testing Process	Charging	Discharging	Charging	Discharging	Charging	Discharging
Cell surface	86.1	84.2	87.1	84.4	Ref	Ref
PCB near U1	68.5	73.3	67.8	73.5	130-(45-25)=110	130-(55-25.0)=100
PCB near MOSFET (Q1)	63.4	68.3	62.7	68.1	130-(45-25)=110	130-(55-25.0)=100
Plastic Enclosure	51.3	53. 2	52.5	53.9	80-(45-25)=60	80-(55-25.0)=50
Ambient	25.0	25.0	25.0	25.0	--	--

Attachment – Photo Documentation



Fig.1



Fig.2

****END OF THE REPORT****